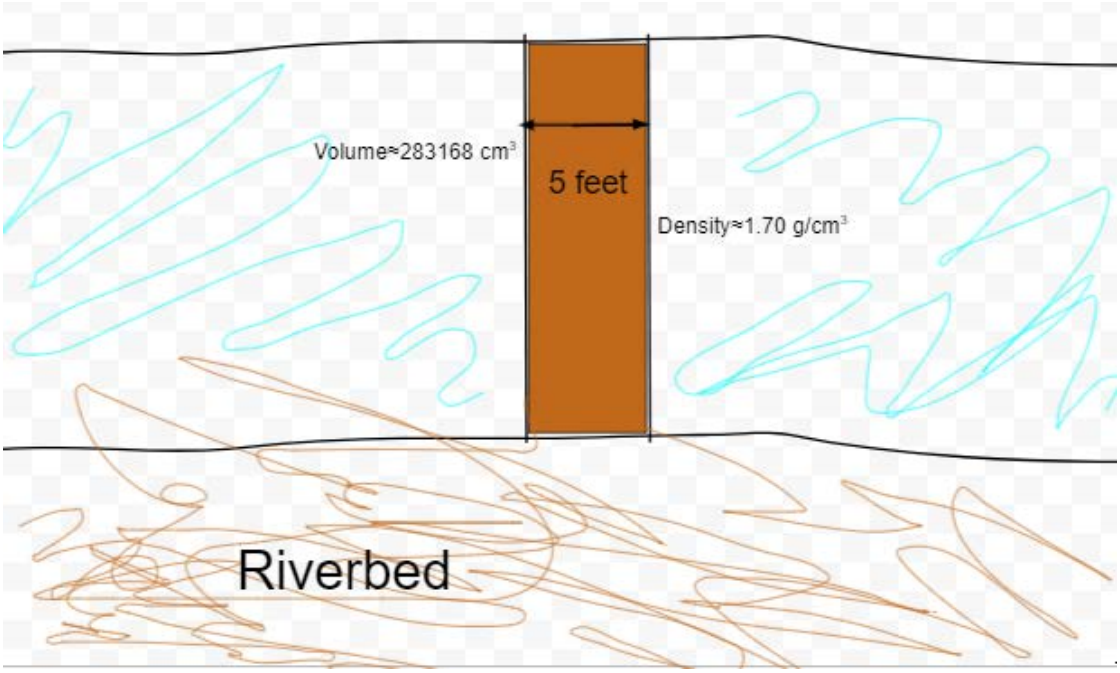
Dynamic Planet SSSS 2021 test (Thenerdicat)

Total points if all questions are answered correctly: 231

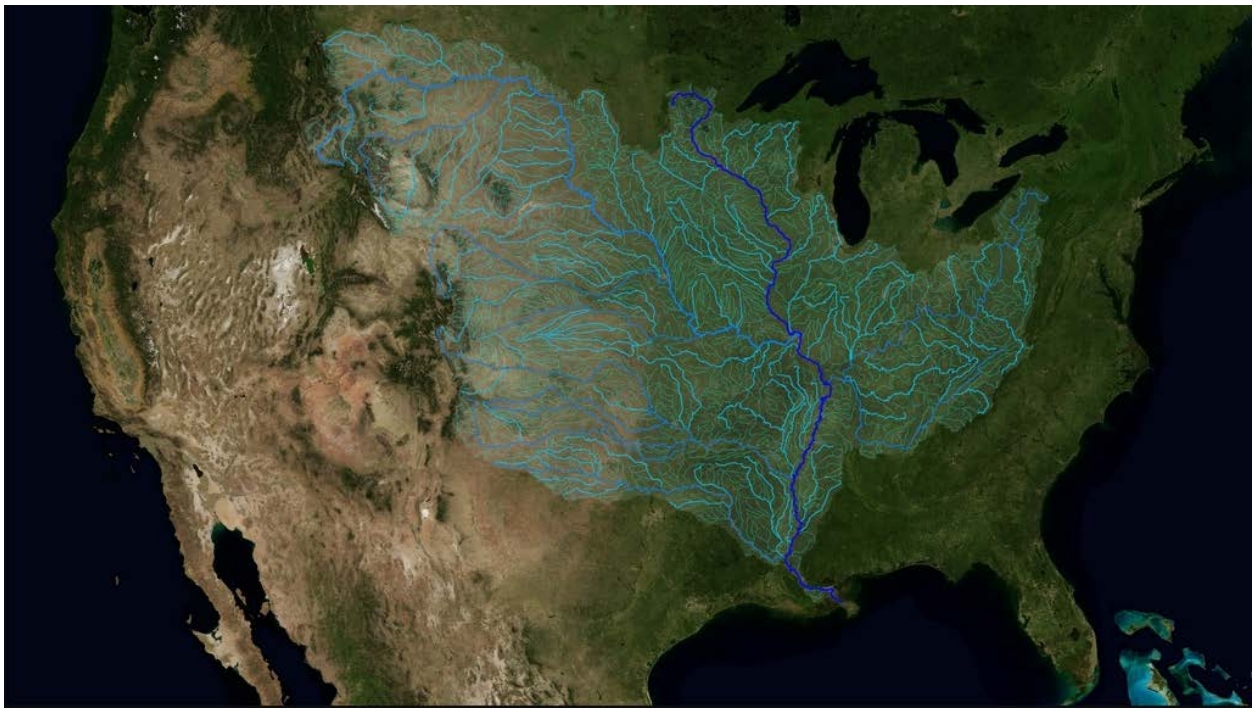
Directions: Points are scored based off what parts of the question you have answered correctly, and long answer problems are worth more in point value as they are long answer. It is better to list as much information for a question, but it must be relevant. Points are taken away if a certain part of an answer is incorrect, and an appropriate amount of points will be taken off corresponding to how important the incorrect part of the answer is. Good luck!

1

1. (1 point) Freshwater has the salinity of less than ___ ppm
2. (8 points) How has sediment transport been impacted by human activity? Explain.
    a. (5 points) Explain how humans have contributed to an excess amount of sediment in watersheds
    b. (3 points) Explain how this affects water quality (things such as factors that influence water quality, etc.)
3. (10 points) How do dams affect local watersheds? Explain.
    a. (5 points) Explain the things that change in watersheds when artificial dams are built
    b. (5 points) Explain how this affects local ecosystems
4. (2 points) What is the average speed of sediment flow (per hour) based off the diagram below if the speed it took for the sediment to get transported was 10 hours? Express in grams/hour



5. (2 points) Below is a map of the Mississippi River drainage basin. What type of stream pattern does this follow?



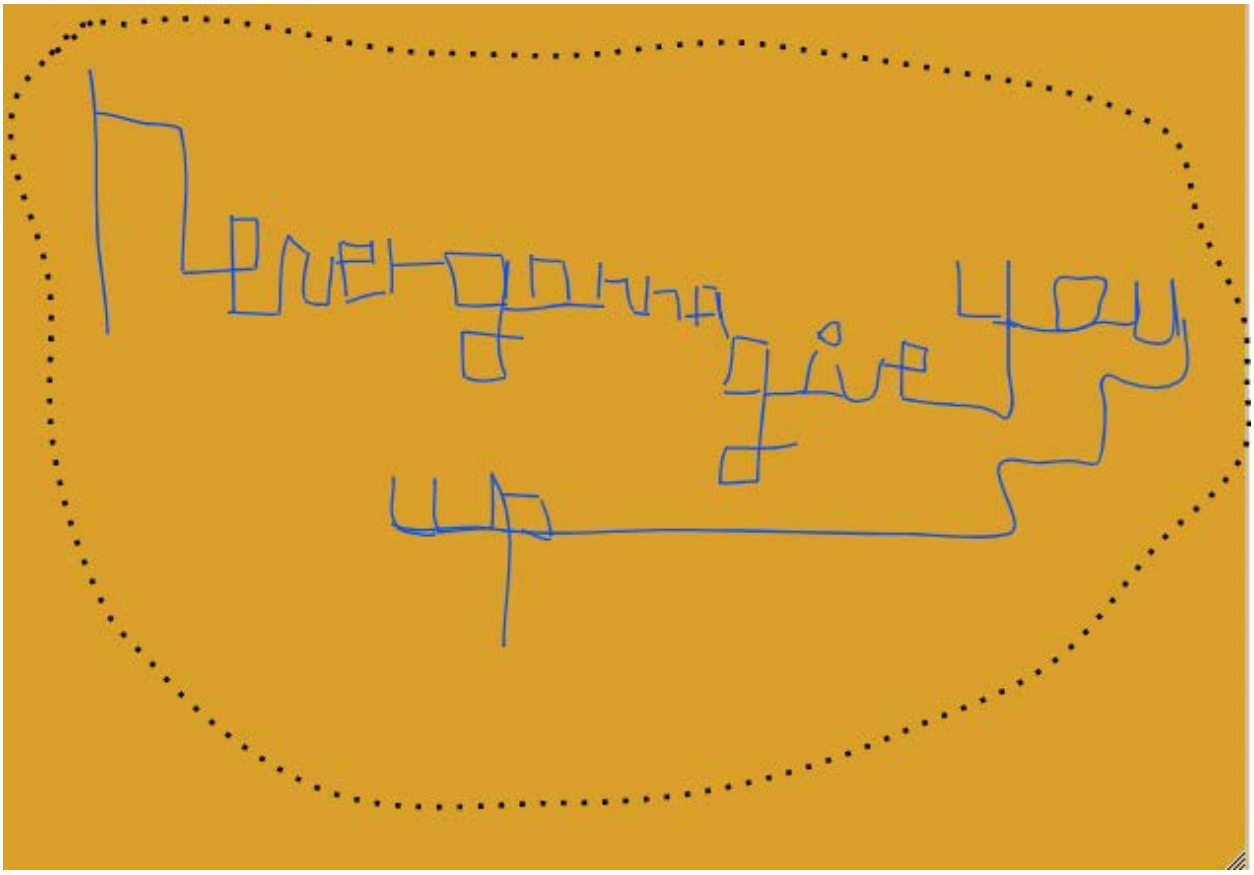
6. (7 points) List the steps of watershed eutrophication. Explain how watersheds get an excess of nutrients (4 points) and then explain how watersheds "die out" from the excess of nutrients (3 points)
7. (1 point) Lake Baikal's depths were first majorly explored in 1989 by a group of scientists from National Geographic Society & the Soviet Academy of Sciences. The life found at the bottom of the lake most closely resembles what type of ecosystem?
    A) Benthic life on the ocean floor (think benthic sediments)
    B) Life in hydrothermal vents (black and white smokers)
    C) Life in a coral reef (shallow waters, plenty of plant and animal life)
    D) Life in heavily saline lakes (Dead Sea as reference)
8. (1 point) Which of these organisms <u>do not</u> live in a freshwater ecosystem?
    A) Mayfly
    B) Asian Carp
    C) Water hyacinth
    D) Mayfly
    E) Zooxanthellae
    F) Water lettuce
9. (5 points) (2 points) Are watersheds like rivers and lakes part of the water table? (3 points) Why or why not?
10. (3 points) What are the general steps of meander formation in streams? (2 points) How common are they? (1 point)
11. (6 points) What are the most common water pollutants in freshwater rivers worldwide?
12. (2 points) Ox-bow lakes are formed from:

A) Meander scars
B) Floodplains
C) Meanders
D) Deltas

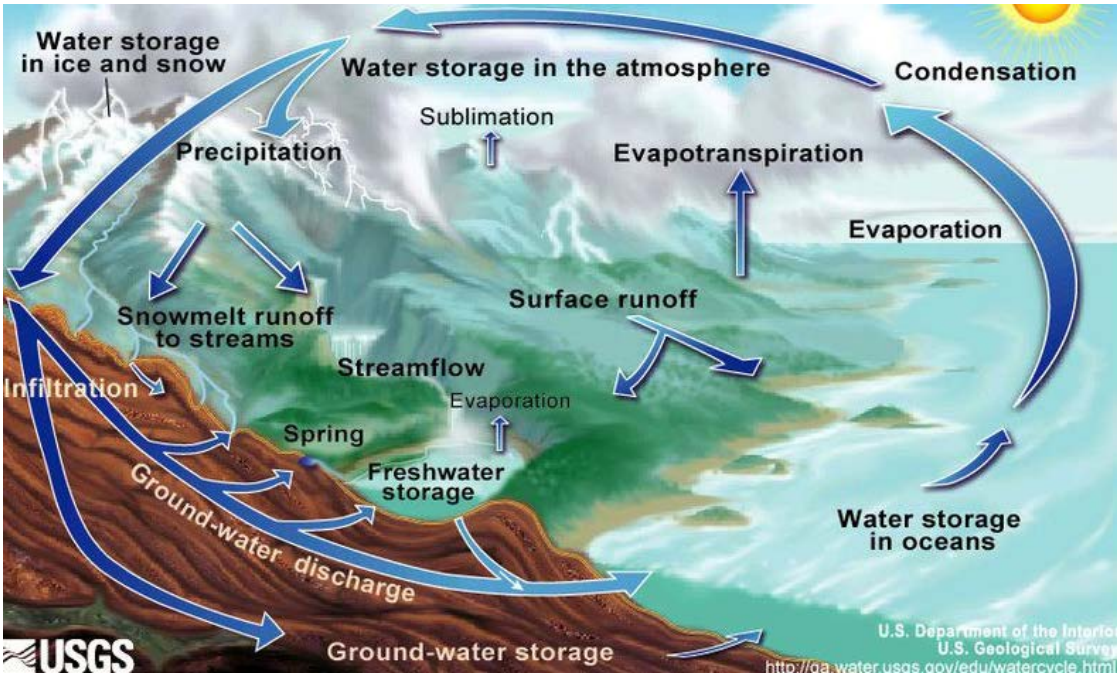
13. (2 points) Observe the photo below. Identify the type of wetland.



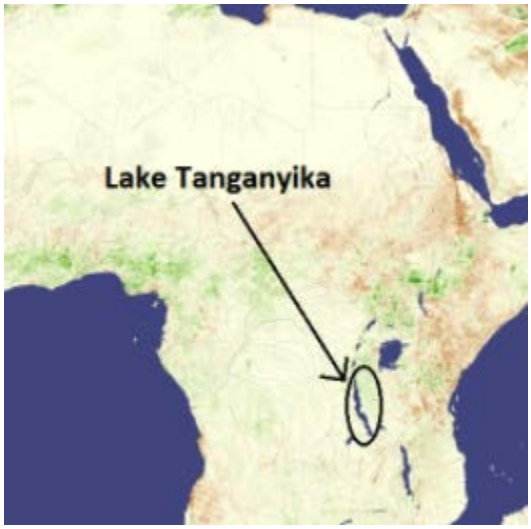
14. (2 points) Which drainage pattern is the most common?
15. (2 points) Which factor influences the quality of drinking water **the most**?
    A) Temperature
    B) pH
    C) Dissolved oxygen
    D) TDS
16. (2 points) What drainage pattern is the one pictured below? (The drainage pattern is in blue)



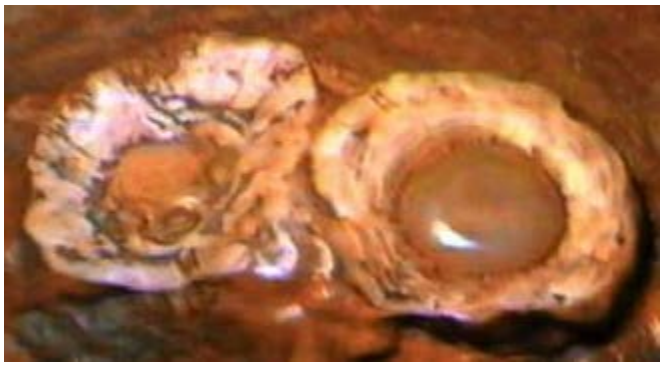
17. (3 points) What is the main difference between a braided stream channel & an anastomose channel?
18. (10 points) Explain the main steps for lake formation. (2 points) List out all 21 types of the lakes and explain how they form. (8 points)
19. (2 points) Which cycle is being depicted below?



20. (2 points) Peat bogs are a ___ sink over thousands of years
21. (2 points) Glaciers and ice caps store __% of all water on Earth
22. (6 points) How are waterfalls generally formed? Explain
23. (6 points) How did this lake below form? Explain



24. (3 points) Why is sediment budget so difficult to calculate properly?
25. (5 points) How has the Nile River's ability to provide water for daily human life been threatened? Hint: Human activities have caused the river to have poor water quality.
26. (2 points) Calculate the porosity of a cube of clay with the ratio of air to total volume as 4:16. Express as a percentage
27. (2 points) What is considered a fast velocity of groundwater? (In centimeters/day)
28. (4 points) Order these materials from lowest to highest average porosity: salts, gravel, silt, clay, sand
29. (6 points) What does Darcy's law state?
30. (1 point) _____ erodes from acid rainwater, forming karst formations. (Hint, usually in the form of calcite or aragonite.
31. (3 points) "Fried eggs" is a famous geological feature found in Luray Caverns. How did they form? (photo below)



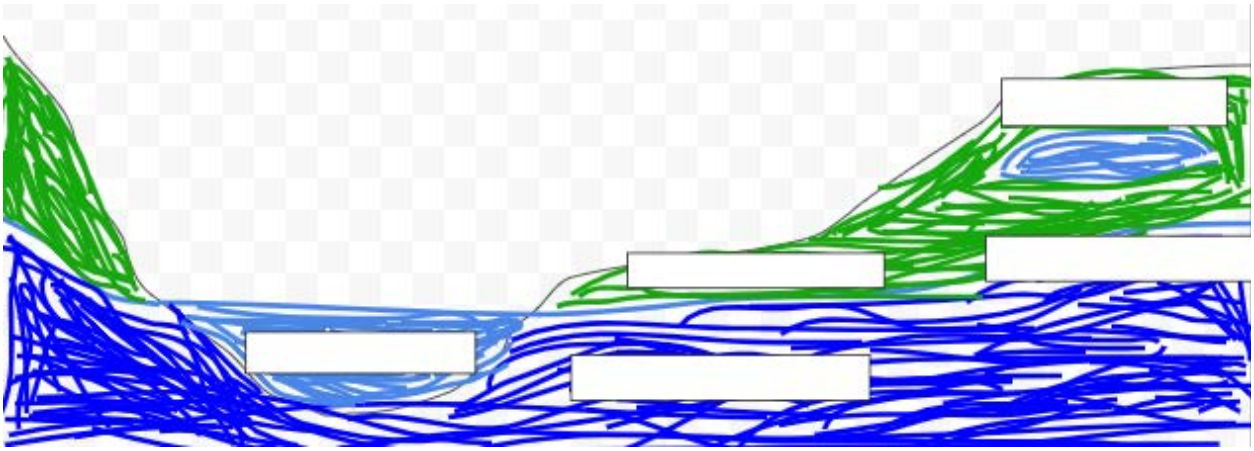
32. (5 points) Compare and contrast stalagmites and stalactites.

33. (2 points) Find the vertical hydraulic gradient of a stream when the head difference taken from Point A to Point B is 13 meters, and the elevation difference is 7 meters. Round to the nearest thousandth (no fractions).
34. (3 points) Is clay a common material found in calderas? Explain why or not.
35. (4 points) What is the process that water takes when it becomes surface runoff?
36. (4 points) Explain how karst formations form.
37. (3 points) A pH of ____ or lower generally indicates poor water quality
38. (4 points) How do sinkholes generally form?
39. (5 points) Why are geological features such as aquifers, aquitards, the water table, etc. important? What role do they have to play in ecosystems?
40. (3 points) What type of watershed is pictured below? How does it form and where is it generally found? (Be specific)



41. (5 points) In what ways is groundwater filtered? Why is it generally cleaner than surface water? How much cleaner?
42. (5 points) Is clay's porosity higher than silt? Why or why not? Explain.
43. (3 points) How much arsenic in drinking water is safe? What is a "safe", government-issued level of arsenic in drinking water? (Hint, EPA)
44. (6 points) Why is salt in seawater bad for drinking? What is a safe level of dissolved salt in drinking water?
45. (2 points) Why are seasonal lakes named ephemeral lakes?
46. (2 points) What color of Secchi disks are used to measure freshwater turbidity?

47. (6 points) How does xeriscaping help watersheds?
48. (5 points) Why is Asian Carp so invasive to the Mississippi River and its tributaries?
49. (5 points) What role does the Chesapeake Bay watershed play in its ecosystem and the surrounding states?
50. (4 points) Why are cattails so important in preserving watersheds? Select all that apply
    A) They provide a habitat for organisms like birds to live among
    B) They soften heavy weather
    C) They filter runoff water, keeping watersheds clean
    D) They protect pond banks from erosion & reduce waves and winds hitting the shore
51. (4 points) How are droughts typically predicted? Does the water table change when an area undergoes drought?
52. (2 points) Explain the relationship between aquifers and groundwater
53. (5 points) How does groundwater drilling affect the land and local water tables? Is this ethical for the environment?
54. (2 points) An aquifer is usually found _____ the water table
55. (5 points) How does land subsidence affect groundwater and the water table?
56. (6 points) What are some ways that humans impact groundwater quality?
    a. (3 points) Explain the actions of humans that impact groundwater
    b. (3 points) Explain the effects this has on groundwater quality
57. (4 points) How do pollutants such as arsenic and antimony get into drinking water?
58. (2 points) On average, how many meters below ground is the unsaturated zone?
59. (3 points) How is the water balance equation an important part of hydrology?
60. (10 points) Label the diagram below



SSSS - Repaired practice edition

Division B | Dynamic Planet | Repair appendix

This appendix contains authored replacements and supplied written alternatives. It does not reconstruct unpublished historical material. For the listed item IDs, use this appendix instead of the archived wording and key.

Name: _____ Date: _____

repair-case. (2 points) Authored written substitute for unavailable source material: Every lake mixes twice per year?

Unit: Lakes, wetlands, and freshwater storage.
Worked-case data: A storage feature lowers peak outflow but releases the same event volume later. Was water destroyed?

A. No

B. Always

C. Only if deep

D. Only if freshwater

Answer: _____